Question 1:

Expression Evaluation with Short-Circuit Logic Evaluate each of the following expressions if a is 6, b is 9, c is 14, and flag is 1. Which parts of these expressions are not evaluated due to short-circuit evaluation?

1. c == a + b || !flag

2. a != 7 && flag || c >= 6

3. !(b <= 12) && a % 2 == 0

4. !(a > 5 || c < a + b)

Answers:

a=6 b=9 c=14 flag=1

1. c == a+b || !flag

c == a+b

14 == 6+9

14 == 15 → false (0)

! flag → !1 → 0

- Answer = 0

2. a != 7 && flag || c >= 6

a != 7 → 6 != 7 → 1

1 && 1 → 1

c >= 6 → 14 >= 6 → 1

- Answer = 1

3. !(b <= 12) && a % 2 == 0

b <= 12 → 9 <= 12 → 1

!(1) → 0

a % 2 == 0 → 6 % 2 == 0 → 1

0 && 1 → 0

- Answer = 0

4. !(a > 5 || c < a + b)

   a > 5 → 6 > 5 → 1

   c < a + b → 14 < 15 → 1

   1 || 1 → 1

   !(1) → 0

   - Answer = 0